

01 Individual Work

PolyBand

Traditional instruments often require too much time, cost, and learning commitment for beginners.

PolyBand is a modular smart instrument system combining string, wind, and percussion into one guided, beginner-friendly music experience.

Product Strategy # Commercial Strategy
Product Definition # User Insight



The Opportunity - User

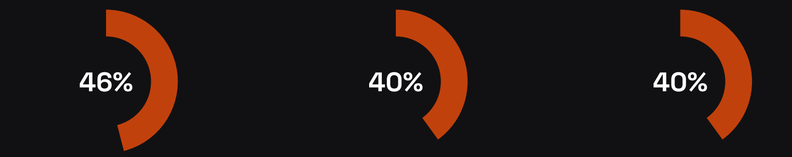
High Entry Barriers Stop Music Enthusiasts Before They Even Begin

Our research revealed that while interest in making music is high, traditional instruments demand too much time, money, and learning commitment.

Research Snapshot

287 Valid Responses Collected from Online Surveys and Interviews

Lack Practice Time | **High Price** | **Difficult to Learn**



46% cited reason for not continuing

40% Find these instruments expensive

40% Find hard to learn instruments

The problem was not a lack of interest in music, but the high commitment required by traditional instruments.

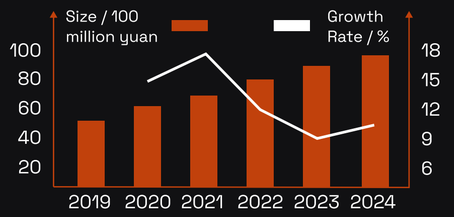


The Opportunity - Market

Smart Music is Gaining Momentum

Music is becoming more accessible, social, and digital. PolyBand combines smart hardware with engaging experiences in a growing market.

Smart Music Market Size



+92%

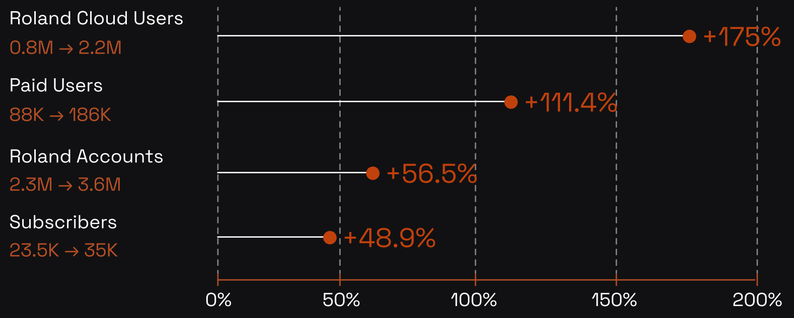
Market Growth

The smart music market nearly doubled from RMB 5.0B in 2019 to RMB 9.6B in 2024.

Digital Growth Beyond Subscription

Growth across Roland's digital ecosystem, 2022-2025

Industry Benchmark: Roland's Digital Ecosystem



Market Takeaways

Growing Market	The smart music market continues to expand.
Digital Engagement	Digital ecosystem usage is growing quickly.
Product Opportunity	Engagement is built before monetization.

Product Strategy

How might we help music beginners explore and perform different instruments with less time, cost,and learning commitment?

Who - Music Curious Beginners

Young users who want to participate in music-making but have limited training, time and budget.



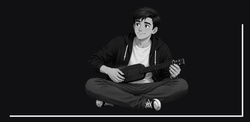


Why - High Barrier Blocks

Traditional instruments require significant time, cost and learning effort before beginners can experience meaningful performance.

What - Modular Smart Instrument

A connected hardware–software system that combines string, wind and percussion experiences in one accessible product.





When - Music-Making Moments

During personal exploration, casual practice, social gatherings and spontaneous performances.

Where - Flexible Everyday Settings

At home, on campus, in community spaces or during informal social activities.



"Combine intuitive hardware and digital guidance to help beginners play and stay engaged."

Guided interaction and connected support let beginners play sooner, explore more, and enjoy music-making with less commitment.



Product Definition

One System, Three Ways to Play

PolyBand is a modular smart instrument system that combines string, wind, and percussion experiences through connected hardware and digital guidance.

Core Value

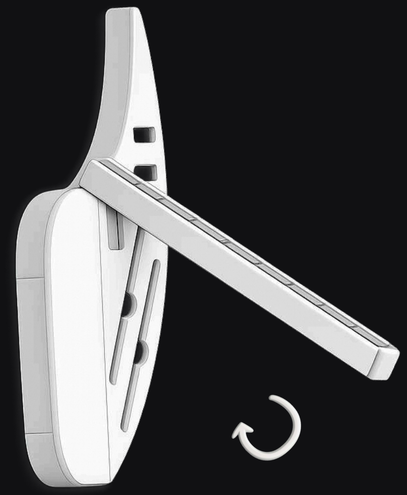
1 Modular — Three instruments in one system.

2 Guided — Visual cues simplify first-time play.

3 Portable — Fold, detach and carry anywhere.



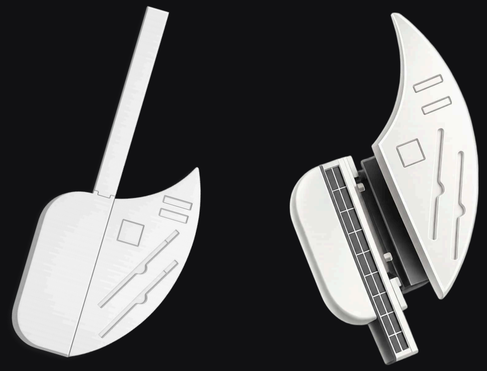
Fold, Detach and Carry



- Foldable and Detachable Compact Storage
- Club and Classroom Ready Portable Performance
- Backpack-Friendly Lightweight Music Experience

String Module

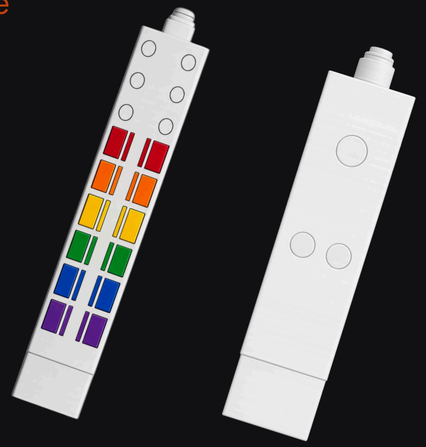
A foldable, stringless instrument that simplifies chords and strumming through guided button-and-touch interactions. It also serves as PolyBand's main body and built-in speaker.



- Stringless Interaction
- Guided Chords
- Foldable Design
- Built-in Speaker

Wind Module

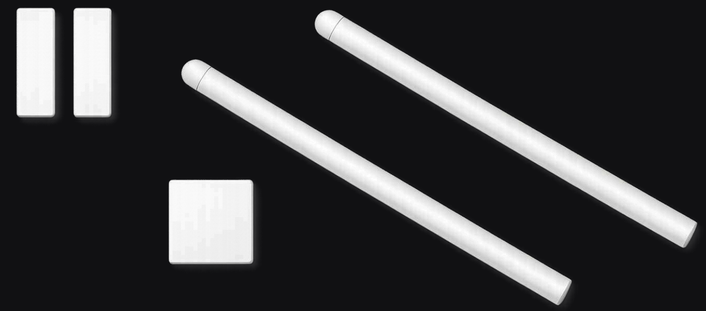
A vertically played electronic wind instrument with visible, color-coded controls. Pressure and airflow sensing lower the entry barrier while preserving expressive breath control.



- Color-Coded Fingering
- Vertical Playing
- Pressure Sensing
- Airflow Control

Percussion Module

Two drumsticks work with three locators and the companion app to create a virtual percussion zone. It can also provide rhythmic accompaniment for the other modules.



- Motion Sticks
- Spatial Locators
- Virtual Drum Zone
- Auto Rhythm

Connected Playing Experience

One App Connects Three Ways to Play

PolyBand's companion app connects all three modules into one guided music experience - from device setup and song selection to real-time playing and accompaniment.

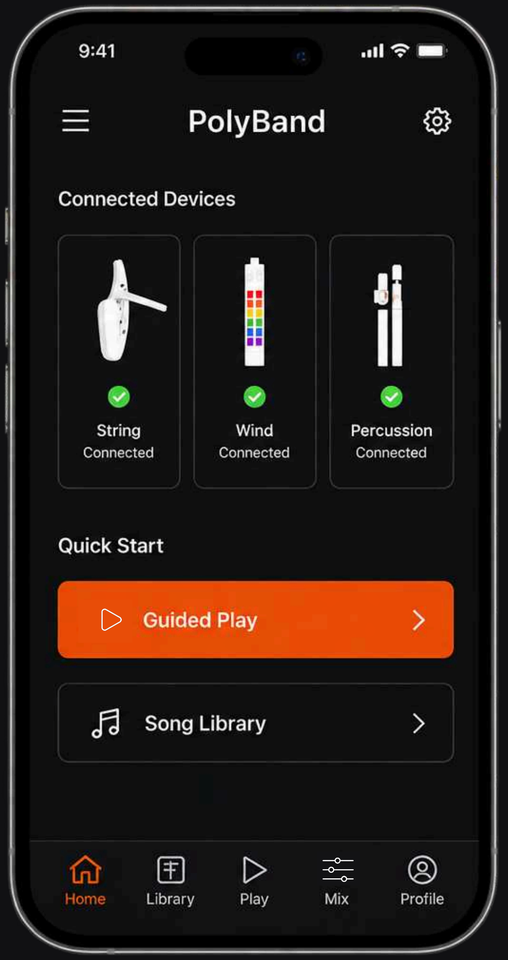
Hardware Inputs

String Module
Select + Strum

Wind Module
Press + Blow

Percussion Module
Locate + Strike











Experience Output

Instant Sound
Immediate feedback while playing.

Multi-Instrument Mix
Combine string, windand percussion.

Solo Play
Others can provide accompaniment.

Group Play
Play alone or with others.

Hardware Input
Play through any or all three modules.

Polyband APP
Receives input, guides play and controls mix.

Guided Sound
Real-time feedback and auto accombaniment.

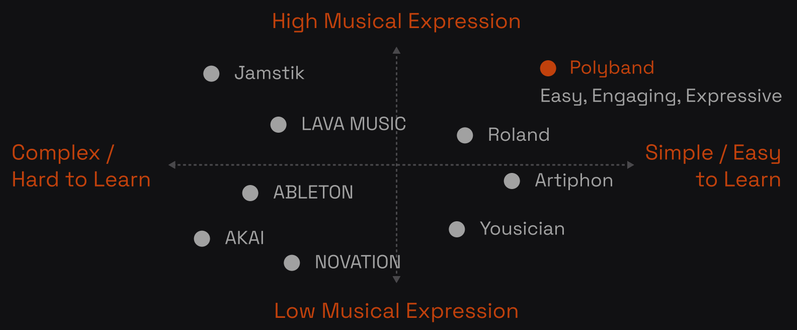
Solo/Combined Performance
Play solo or together for a complete music experience.

Market Fit

From Priority to Release Scope

PolyBand combines a low learning barrier with an integrated multi-instrument experience for easier, more flexible music exploration.

Market Positioning



Conclusion

PolyBand combines a **low learning barrier** with an **integrated multi-instrument experience**.

Target Market

Start with **music-curious beginners**, then expand toward **social creation** and **institutional use**.

Level 01 — Primary

Music-Curious Beginners

First-time players looking for an **approachable**, guided way to explore music and **build confidence**.

Low Barrier · First-Time Play · Guided Learning

Level 02 — Secondary

Casual Creators & Social Players

Users who enjoy **creating**, **sharing**, and playing music in **casual social settings**.

Fun Sharing · Light Performance · Social Play

Level 03 — Expansion

Education & Community

Institutions and communities integrating PolyBand into **learning**, **programs**, and **events**.

Classroom Use · Workshops · Community Events

Business Viability

SWOT Analysis

S
- Hardware + digital guidance
- Modular experience
- **Low learning barrier**

W
- **Integration complexity**
- Limited prototype maturity

Poly band

O
- **Accessible music creation**
- Education & social scenarios

T
- **Established brands**
- Supply-chain uncertainty

Business Model

PolyBand combines hardware sales, digital services, and institutional use to create multiple revenue streams.

Hardware Sales

Sell PolyBand devices as the **entry product**.

Subscription

Unlock **premium content** and cloud tools.

Education / B2B

Workshops and **institutional programs**.

Pricing

Pricing lowers the entry barrier while encouraging higher-value bundles and recurring services.

01 Entry > **02 Bundle** > **03 Recurring Value**

Lower upfront cost to get started easily.

Value-packed sets to enhance the experience.

Ongoing content for long-term value.

Market Execution

Go-to-Market

PolyBand enters through low-barrier trials, creator exposure, and guided first-use experiences.

1 Launch Entry
- Campus Clubs
- Beginner Communities
- Workshop Trials
- Creator Seeding

2 Awareness
- Short-form Video
- Creator Collaborations
- Demo Performances
- Sharing Moments

3 Conversion
- Try Before Buying
- Starter Bundle
- Preorder / Limited
- Guided Onboarding

4 Retention Expansion
- App-guided Content
- Song Updates
- Community Challenges
- Add-on Scenarios

Competitive Moat

PolyBand builds defensibility through modular design, guided interaction, and a connected content ecosystem.

Product Moat
- Modular Multi-instrument Design
- Low Learning Barrier
- Connected Hardware-software

Experience Moat
- Guided First-use Journey
- Real-time Feedback
- Complete Beginner Performance

PolyBand

Ecosystem Moat
- Expandable Content System
- App-based Guidance
- Multi-scenario Use

Community Moat
- Social Play Scenarios
- Classroom and Workshop Adoption
- Creator-friendly Sharing Moments

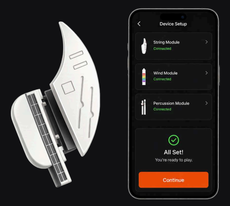
MVP Scope

The MVP Prioritizes the Essential Journey Needed to Validate PolyBand's Core Value

Instead of building every possible feature, the MVP focuses on the essential journey that allows beginners to connect the device, follow guidance, and achieve a satisfying first performance with minimal barriers.

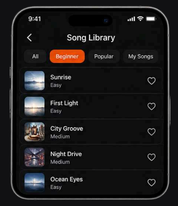
Core MVP Journey

1 Connect — Device Connection



Connect PolyBand modules in seconds

2 Choose — Song Selection



Pick a song you like from our library.

3 Play — Guided Play



Follow visual cues and play with ease

4 Hear — Real-Time Sound



Hear performance in real time.

5 Complete — Basic Accompaniment



Enjoy a complete performance with automatic accompaniment.

Priority Matrix

- MVP – Essential
- MVP – Stretch
- Later Iterations



From Priority to Release Scope

The MVP keeps only the features required to deliver the core beginner experience, while more complex capabilities are reserved for later iterations.

Release Scope

- Device Connection
- Song Selection
- Guided Play
- Real-Time Sound

→ **Required to Complete the Core Journey**

These features enable beginners to start quickly, follow guidance,and hear their performance in real time.

- Basic Accompaniment → **Experience Boost**

Enriches solo play with auto accompaniment.

- Group Play → **Social Expansion**

Deferred to later iterations after core validation.

Key Risks to Validate

1 Connection Reliability

Validate whether beginners can connect and prepare the system without external assistance smoothly and independently.

2 Guidance Clarity

Validate whether visual guidance reduces learning barriers and helps users perform without prior training.

3 Performance Completeness

Validate whether combined feedback and accompaniment help beginners feel successful after their first performance.

Prototype Feedback

We invited teammates to try the physical mockup and simulate a collaborative music-making scenario.



“The core interaction is intuitive and engaging.”

“Real-time feedback and ergonomics still need refinement.”

Key Insights	Areas to Improve
• Intuitive to explore	• Clearer real-time feedback
• Engaging multi-instrument play	• Refine grip & ergonomics
• Low first-use barrier	• Improve interaction consistency

Broader Impact

Beyond its core product experience, PolyBand can create broader value across education and cultural engagement. Its accessible, modular design can support learning communities, while public experiences and collaborations help reach wider audiences and encourage active participation in music.

Education Value



Interdisciplinary Learning	Music × Design × Technology
Accessible Music Education	Lower barriers for classrooms and student clubs
Hands-on Creativity	Encourage learning through making and exploration

Social Reach





Online Engagement	Short-form content and creator participation
Offline Experience	Public demos, workshops and campus events
Cultural Collaboration	Partnerships with cultural IPs, brands or institutions

Risk & Mitigation

Three key risks must be reduced before PolyBand can successfully move beyond the prototype stage.

	Key Risk	Mitigation
01	**Demand Uncertainty** Market acceptance is still unproven.	**Small-scale Validation** Test demand through campus pilots and pre-orders.
02	**Technical Complexity** Multiple sensing systems increase integration difficulty.	**Modular Development** Validate core modules separately before full integration.
03	**Production Resources** Manufacturing and team resources remain limited.	**Small-batch Development** Use standardized parts and external technical support.
04	**Supply Chain Dependency** Key components may face cost or availability fluctuations.	**Flexible Sourcing** Use standardized parts and multiple supplier options.

Future Iteration

Based on current prototype feedback and identified risks, PolyBand can evolve through three focused stages — validating the core experience, improving product completeness, and expanding into broader collaborative use.

Phase 01 Validate

Refine the Core Experience

- Test with more beginner users
- Improve interaction feedback
- Optimize grip and ergonomics
- Validate device connection stability

Confirm that beginners can complete a satisfying first music experience with minimal guidance.

→

Phase 02 Improve

Build a More Complete Product

- Expand song content
- Improve guided play
- Refine auto accompaniment
- Optimize multi-module integration

Strengthen the experience from basic playability to a more complete performance journey.

→

Phase 03 Expand

Extend the PolyBand Ecosystem

- Enable group play
- Support community activities
- Develop educational applications
- Explore cultural and brand collaborations

Extend PolyBand into a connected platform for collaborative play, learning, and cultural engagement.